

EXECUTIVE BRIEF

Reducing Dimensions Without Losing What Matters

Decision controls for SVD and PCA

SRAI - Statistics, Reasoning and Artificial Intelligence

Book 1: Mathematical Foundations | Lesson 7

Executive issue

SVD and PCA can turn many correlated indicators into a smaller set of reproducible dimensions. This can improve clarity, speed and model stability. It can also remove low-variance information that is operationally, ethically or politically important.

What leaders should require

- A named decision purpose and accountable owner.
- A documented choice of variables, units, centering and scaling.
- A transparent rule for the number of retained dimensions.
- Reconstruction and downstream-performance evidence, including subgroup checks.
- A preservation-and-loss statement in plain language.
- Access to original variables, residuals and an escalation or reversal path.

Five questions for the decision meeting

- What decision will use the compressed representation?
- What information does it preserve, and according to which metric?
- Which groups, anomalies or rare events are least well represented?
- How does compression change downstream results and error costs?
- Who can stop, revise or reverse its use?

Interpretation guardrails

Explained variance is not accuracy, causal importance or fairness. Component names are hypotheses, not facts. Opposite component signs can be mathematically equivalent. A technically optimal low-rank approximation can still be institutionally unacceptable.

Recommended decision condition

Authorize dimensionality reduction only for a bounded use where preprocessing is reproducible, loss is measured, affected groups are reviewed, original information remains recoverable and a named owner can revise or stop the approach.

Bottom line

Compression is a decision about information removal. Treat it as a governed analytical intervention, not merely a preprocessing convenience.